Fall Sem. 2025-26

Tutorial– 1

Programme Name & Branch: B. Tech.

Course Name & Code: Complex Variables and Linear Algebra (BMAT201L)

Slot:

Practice all the questions

1. Examine the analyticity of the following functions and find its derivatives.

   (i) $f(z) = e^z$ (ii) $f(z) = \cos z$

   Hint: ((ii)

$$\begin{aligned}\cos z &= \cos(x+iy)\\ &= \cos x\cos(iy) - \sin x\sin(iy)\\ &= \cos x(\cosh y) - \sin x(i\sinh y)\end{aligned}$$

2. Show that the function $f(z) = \sqrt{|x\,y|}$ satisfies C-R equations at the origin but $f'(z)$ does not exist at the origin.

   Hint: At the origin, verify the derivative of $f(z)$ along the path $y = mx$.

3. Show that the function $f(z) = \begin{cases} \dfrac{x^3(1+i) - y^3(1-i)}{x^2+y^2}, & if \quad z \neq 0 \\ 0, & if \quad z = 0 \end{cases}$

   is not analytic at origin although C-R equations are satisfied at the origin.

   Hint: At the origin, verify the derivative of $f(z)$ along the path $y = mx$.

4. If $f(z) = \dfrac{1}{2}\log(x^2+y^2) + i\tan^{-1}\left(\dfrac{kx}{y}\right)$ is analytic function, then find the values of k .

Answer: $k = -1$

5. If $f(z)$ is analytic function, show that $\left(\dfrac{\partial^2}{\partial x^2} + \dfrac{\partial^2}{\partial y^2}\right)|f(z)|^2 = 4|f'(z)|^2$.

6. If $f(z) = \phi + i\psi$ represents the complex potential function for an electric field and $\psi(x,\ y) = 3x^2y - y^3$, then find $\phi(x,\ y)$.

Answer: $\phi(x,\ y) = x^3 - 3xy^2$

7. If $f(z) = u + iv$ is an analytic function of z and $u - v = (x - y)(x^2 + 4xy + y^2)$, then find $f(z)$ in terms of z.

Answer: $f(z) = -iz^3 + c$.